

Class 02 — Optional Resources

Bong Study Hub — Foundation Batch 2026 · How Machines Learn

Everything you need for Class 02 is already in your **Student Notes**, the **class recording**, and the in-class activities. These resources are **optional** if you want to explore further — nothing here is required, and nothing here will be tested.

Start Here

Before looking at anything external, make sure you can already:

- Explain why hand-written rules can become hard to manage.
- Say the pipeline: DATA → LEARNING → MODEL → PREDICTION .
- Explain why a prediction isn't a guarantee.

If any of those feel shaky, your **Student Notes** (02_Student_Notes/) are the fastest way to fix that — faster than any external link below.

Optional Exploration

Five small, hand-picked resources. **No more than two are marked Recommended** — the rest are genuinely optional, pick only what interests you.

Resource 1 — Teachable Machine

- **Source:** Google Creative Lab
- **Type:** Free interactive tool (browser-based, no account needed)
- **Time:** ~15 minutes
- **Level:** Beginner — no coding required
- **Priority:** ★ RECOMMENDED
- **Link:** <https://teachablemachine.withgoogle.com/>

Why look at it: This lets you *actually do* what we described in class — give a computer examples (photos, sounds, or poses), and watch it build something from them that can then guess on something new. It's the DATA → LEARNING → MODEL → PREDICTION pipeline, hands-on, in your browser.

Focus on: Training a simple image classifier with your webcam (e.g., two hand gestures) and noticing how the model's guess changes as you give it more or different examples.

Skip: The "export your model" / website-integration features — that's a later, technical step, not needed here.

Resource 2 — AI for Oceans

- **Source:** Code.org (nonprofit, official)
- **Type:** Free interactive activity
- **Time:** ~15–20 minutes
- **Level:** Beginner — no coding required
- **Priority:** ★ RECOMMENDED
- **Link:** <https://code.org/oceans>

Why look at it: You train a system to sort "fish" from "trash" by giving it examples — then you can deliberately give it bad or narrow examples and watch it make worse guesses. It's a very close hands-on match to this class's cats-vs-dogs reasoning and the data-quality idea ("poor data → poor predictions").

Focus on: What happens to the system's guesses when you feed it narrow or one-sided examples on purpose.

Skip: Nothing technical to skip — it's built for beginners throughout. (It does use the word "bias" briefly; that's not a term this class requires, but it's fine to notice it here.)

Resource 3 — How Netflix's Recommendations System Works

- **Source:** Netflix Help Center (official)
- **Type:** Short article
- **Time:** ~5–8 minutes
- **Level:** Beginner
- **Priority:** Optional
- **Link:** <https://help.netflix.com/en/node/100639>

Why look at it: This is Netflix's own plain-language explanation of its recommendation system — a real-world version of the "AI Around Us" examples from class, described conceptually rather than technically.

Focus on: How it uses your viewing history and patterns across similar viewers — the "examples in, best guess out" shape.

Skip: Nothing — the whole page stays at a conceptual level.

Resource 4 — Google Maps 101: How AI Helps Predict Traffic and Determine Routes

- **Source:** Google (official blog)
- **Type:** Short article
- **Time:** ~8 minutes
- **Level:** Beginner
- **Priority:** Optional
- **Link:** <https://blog.google/products/maps/google-maps-101-how-ai-helps-predict-traffic-and-determine-routes/>

Why look at it: A plain-language look at how Maps combines historical traffic patterns with live data to predict your ETA — a real-world "AI Around Us" example, in the company's own words.

Focus on: The idea of learning from historical patterns *plus* current data to make a prediction.

Skip: The single brief mention of "Graph Neural Networks" — that's a later, technical topic, not needed for this class.

Resource 5 — AI vs. Machine Learning vs. Deep Learning vs. Neural Networks

- **Source:** IBM Think
- **Type:** Short article with diagrams
- **Time:** ~10 minutes
- **Level:** Beginner-to-intermediate
- **Priority:** Optional curiosity — not required for Class 02
- **Link:** <https://www.ibm.com/think/topics/ai-vs-machine-learning-vs-deep-learning-vs-neural-networks>

Why look at it: It draws the same broad-to-narrow relationship we saw in class (AI ⊃ Machine Learning ⊃ Deep Learning) with a different set of everyday examples.

Focus on: The opening explanation and diagram of how the four terms relate.

Skip: The "Neural Networks" section, which briefly mentions weights and backpropagation — that's later curriculum, not Class 02.

You Will Meet These Later

You may come across these words while exploring AI online. **You do not need to learn them yet.** We will build toward them step by step, the same way this class built toward Machine Learning from Class 01's foundations:

Programming → Computer Science → Mathematics → Data → Machine Learning → Deep Learning → LLMs → Agents

Seeing an unfamiliar term in one of these resources (or anywhere else online) is not a sign you're behind — it's a preview of a step further up the staircase.

A Note on How to Use These

Don't try to understand everything. If you open one of these and it starts feeling technical, that's your cue to stop, not a sign you should push through. Before going deeper into anything AI-related online, ask yourself:

- Does this reinforce something from class, or is it introducing a new technical topic?
- Do I already understand the basic idea here, or am I skimming past confusion?

If the answer is "this has gone technical" — stop there. You've gotten what you need from it, and the rest is for a later class.